# Validated Ensembles, Physics-Gated Post-Processing and Selection-Faithful Queue Forecasting for TrafficFlowBench

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***Abstract*—We describe our solution to the 2026 IEEE Big Data Cup *TrafficFlowBench*, a four-task freeway benchmark (state reconstruction, queue forecasting, physical consistency and origin–destination estimation) scored as one weighted total. Our final submissions reach a public-leaderboard score of 0.88281, up from 0.80939 for the public notebook we started from. The Task 1 reconstruction is a residual ensemble of 17 gradient-boosted members (LightGBM, CatBoost, XGBoost) trained on the training months plus *transductive* rows from the released cells of the target months. A dedicated specialist handles the 90-minute blackouts after Task 2 forecast origins, and a low-rank tensor completion of the month's own flow field adds a further member. Task 3 is improved only through Task 1: an FD-gated reconciliation and a 1-D total-variation smoother move the reconstructed density towards the fundamental diagram and remove noise that breaks conservation. For Task 2 we found that the official windows are selected with a persistence cap evaluated on the *full* state. Training the ongoing-queue model on windows selected the same way gave our largest single gain after the first model. Onset queues are forecast at $T{+}30$ only, with a CatBoost classifier and an expected-IoU decoder. Task 4 uses the organizer's prior-regularised NNLS. All changes passed through day-blocked cross-validation, a pseudo-March evaluation and pre-registered gates. Every Task 2 forecast uses only data stamped at or before its origin $T$. We report the full progression, the offline-to-leaderboard transfer and a list of negative results.**

## I. Introduction

TrafficFlowBench [1] asks four questions about ten synthetic but calibrated directional freeway corridors ("panels") at 5-minute resolution: fill in masked detector cells (Task 1), forecast which links are queued in the next 30 minutes (Task 2), be physically consistent while doing Task 1 (Task 3), and recover path flows from link counts (Task 4). The score is

$$S = 0.35\,S_{\text{state}} + 0.30\,S_{\text{queue}} + 0.05\,S_{\text{FD}} + 0.10\,S_{\text{LWR}} + 0.20\,S_{\text{ODME}}, \tag{1}$$

where $S_{\text{physics}} = S_{\text{FD}}/3 + 2S_{\text{LWR}}/3$ carries weight 0.15. The public leaderboard (LB) scores the validation month (March 2031); the private LB scores April 2031. The training split has 273 days with released answers.

**Contributions.** (i) A transductive residual ensemble for Task 1 that also improves Task 3, with an explicit blackout specialist and a month-level low-rank member (Sec. IV). (ii) Two physics-based post-processing steps for Task 3 that can be applied to any reconstruction (Sec. V). (iii) A reconstruction of the official Task 2 window selector, and selection-faithful training that follows from it (Sec. VI). (iv) A validation protocol of day-blocked folds, pseudo-March windows and pre-registered gates. It recorded how offline gains transfer to the LB and helped us avoid tuning to LB noise (Sec. III). (v) A documented list of what did not work (Sec. IX). Fig. 1 summarises the final pipeline.

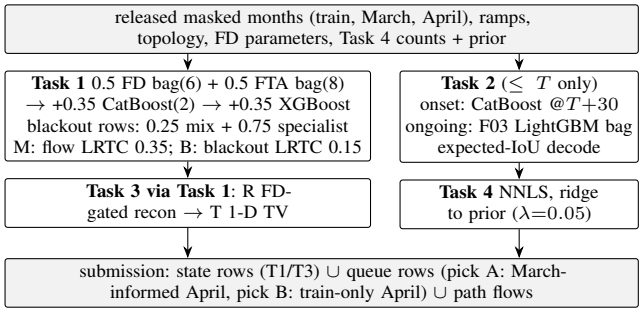



Fig. 1. Final pipeline. The two picks share every state, March queue and path-flow row; they differ only in 200 April queue rows.

Organizer requirements for this report (forum post 744187) are a PDF "describing your method, your analysis and your results across the four tasks". There is no template and no page limit. Every number below is traceable to a file in our repository. Files are named in `typewriter`, and the main log is `experiments/TRACKER.md`. Numbers that are estimates are marked as such.

## II. Problem, Data and Rules

**Task 1 (state).** Every calendar day is masked in one regime (R1/R2/R3 = 20/30/50 % of eligible cells). The submission gives speed and flow for every target cell. $S_{\text{state}} = 0.54\max(0, 1 - \text{RMSE}_v/25) + 0.46\max(0, 1 - \text{RMSE}_{q/\text{lane}}/600)$, averaged over regimes, then panels. **Task 2 (queue).** Each of 8 panels has 5 *onset* and 5 *ongoing* windows per split. Each window gives 60 minutes of history before an origin $T$, and the released row at $T$ may also be used. The target is a binary queue map (speed $\le 0.6\,v_f$ of the noise-free state) over $T{+}5, \ldots, T{+}30$ for every link. The score is a per-window space-time IoU, averaged window → condition → panel. **Task 3 (physics)** is computed from the Task 1 file. $S_{\text{FD}}$ measures the distance to the fundamental diagram (FD). $S_{\text{LWR}}$ measures violations of LWR vehicle conservation [3], $N(t{+}1) - N(t) = \Delta t(q_{\text{in}} + r_{\text{on}} - q_{\text{out}} - r_{\text{off}})$, with $N = (q/\max(v, 1))L$. It uses organizer-held boundary

fluxes, so it cannot be scored locally. **Task 4 (ODME)** asks for path flows given the incidence matrix $A$, link counts $c$ and a weak prior $b$.

**Time-stamp rule.** Task 1 is offline, so any released observation of the split may be used (organizer, post 742068). For Task 2, a forecast with origin $T$ may use released data stamped $\leq T$. That covers the training split, earlier days of the same split and, for April, all of March (posts 742068, 745700, 748789). It may not use ramp values inside the horizon or later mainline observations. At award review the organizers rerun Task 2 with every row after each window's $T$ removed, and the predictions must not change (post 749601). We read that post strictly: every model, threshold and selection choice that affects April rows was made on training and March evidence only.

## III. Validation methodology

### A. Folds and replicas

All Kaggle runs share frozen folds: 3 contiguous day blocks for Task 1 and 4 for Task 2. Task 1 candidates are also scored on a *validation-month internal holdout*. In it, 15 % of the released March cells (seed 7) are hidden in every view and never used as a target. This holdout followed the LB closely early on: for example, the low-rank features gave a predicted $+0.0013$ and an observed $+0.0015$ (`TRACKER`, EXP_017).

For Task 2 we wrote a replica of the official selector. The history is $[T-60, T)$ with $\geq 70\,\%$ eligible cells. A window is *ongoing* if some link has $\geq 2$ queued observations, otherwise *onset*. The horizon must contain a queue, the persistence IoU must be $\leq 0.9$, and picks are greedy with 360-minute spacing. The replica reproduces 39/40 official onset origins on the training split. The first version reproduced only 14/40 *ongoing* origins. Evaluating the persistence cap on a time-filled state instead (linear interpolation along time over gaps up to 36 slots, applied to all cells) reproduces 36/40, and 39/40 within 5 minutes (`analysis/t2lab/README.md`). This filled state is used only to select windows, never as a feature.

### B. Pseudo-windows and pseudo-March

The replica produces *pseudo-windows* on the training split: about 2,000–3,000 "official-like" ongoing windows (greedy restarted every day) and the run-start windows. Models are scored out of fold with the official IoU aggregation and paired bootstraps by window within panel. Since March and April come from a new scenario (redrawn ramp demand and a different set of active bottlenecks; `analysis/CAPACITY_SCENARIO_2026-10-11.md`), we also built *pseudo-March* windows from the released masked March layer. Their labels are observed cells only, and they are a decision aid for scenario-dependent choices. A "many-day" variant uses March days $< 15$ for fitting and days $\geq 15$ for evaluation. It mimics April, which can use all of March.

### C. The Task 1 J proxy and pre-registered gates

From October on, every Task 1/3 candidate was evaluated in a paired lab on 4 panels $\times$ 2 masked months (16 fold views). Released cells are hidden in the official run pattern, and blackouts are simulated where official Task 2 windows would put them. The metric is the score-weighted proxy

$$J = 0.35\, S_{\text{state}} + 0.10\, \widetilde{S}_{\text{LWR}} + 0.05\, S_{\text{FD}}, \tag{2}$$

where $\widetilde{S}_{\text{LWR}} = 1 - \sum |\Delta N_{\text{pred}} - \Delta N_{\text{obs}}| / \sum |\Delta N_{\text{obs}}|$ uses released truth outside the target cells. On D12_I405_S the naive baseline gives a proxy $S_{\text{physics}} \approx 0.39$, against the official 0.355 (`TRACKER`, validation design). Before any result was computed, each round wrote a gate into `analysis/t1lab/t1div_lib.py`, with the time recorded. For a new member the gate is: $\Delta J \geq +0.0003$ after the full post-processing, a 95 % bootstrap CI lower bound $> 0$, $\geq 7/8$ panel-months positive, and a margin of $\geq +0.00015$ over a control that simply adds another LightGBM seed at the same weight. Post-processing and weight changes used $\Delta J \geq +0.0002$. Results that were not gate-eligible (post-hoc) are labelled as such below.

### D. Footprint-matched pseudo-March (FM-PM)

Plain train-OOF and pseudo-March deltas mispredicted the LB effect of ongoing changes (MAE 0.0013–0.0015 in $S$). The FM-PM estimate is label-free on the target month. It decodes both candidate files, finds the official windows whose decision changes, and prices each with the paired pseudo-March delta of lab windows from the same panel, a similar time of day ($\pm 2$ h) and a similar change size. The mean is then scaled by the weight of one window. On the five ongoing LB probes it was evaluated on, FM-PM on the *validation* footprint had MAE 0.0005 (`TRACKER` EXP_T2_XFER; Fig. 4, right). FM-PM was calibrated after these probes, so this MAE is in-sample. Following the strict time-stamp reading above, we never used FM-PM computed on April windows as a binding criterion for April rows.

### E. Leaderboard noise and the keep rule

With 40 windows per condition, one ongoing probe has an LB standard deviation of $\pm 0.0012$–$0.0027$ and one onset probe $\pm 0.003$–$0.005$ (`TRACKER` EXP_T2_XFER). A public delta below $+0.0001$ was never accepted as a gain (e.g. F1 $+0.00002$, J1 $+0.00003$). Most Task 1 increments in our final phase were kept because a gate had passed offline, not because of their LB delta.

## IV. Task 1: state reconstruction

### A. Residual members with transductive rows

Every member predicts *residuals* of speed$/v_f$ and flow per lane with respect to a per-station linear interpolation in time. Features include the interpolation itself, neighbour stations at the same instant, a per-day rank-3/rank-5 SVD reconstruction of the interpolated fields, occupancy, characteristic-kernel ("ASM") averages along $\pm 80$ and $\pm 18$ km/h waves, and, in the newer "FD" family, 20 columns of FD-implied densities and flows of several speed views. Profile features were dropped because the profile-free model was better on 10/10 panels. Training rows come from the training split with simulated

TABLE I
TASK 1 PROGRESSION ON THE VALIDATION-MONTH INTERNAL HOLDOUT (COMPOSITE $S_{\text{STATE}}$; TRACKER).

| Model | Holdout | Public LB |
|---|---|---|
| Temporal-interpolation base (re-implemented) | 0.9141 | – |
| LightGBM residual, profile-free | 0.9307 | 0.84855 |
| + per-day rank-3 SVD features | 0.9343 | +0.0015 |
| + rank-5 / occupancy / cross-day low rank | 0.9350 | – |
| 255 leaves, 900 trees, 2× rows | 0.9374 | – |
| 511 leaves, 1500 trees, 4× rows | 0.9386 | +0.0011 |
| + transductive rows (FT) | 0.9406 | +0.0017 |
| + ASM wave features (FTA) | +0.0005 | +0.0003 |

90-minute gaps. They are augmented with *transductive* rows: in 2–4 rounds, 20 % of the released cells of the validation and private months are hidden and used as targets. This is legal because Task 1 is offline (post 742068). It added $+0.0020$ on the holdout (10/10 panels) and $+0.0017$ on the LB (TRACKER EXP_T1_trans).

### B. The final ensemble

The final Task 1 file of both picks is produced as follows (constants in `consolidation/tfb/config.py`, kernel docstrings in `submissions/76_K9/tfb-k9.py`):

1) $m_0 = 0.5\,\overline{\text{FD}} + 0.5\,\overline{\text{FTA}}$, where FTA is a bag of 8 LightGBM members (seeds 0–7, 2–4 transductive rounds) and FD is a bag of 6 (three FD-feature seeds, a simulated-blackout member and two members with 9 ramp features). All are LightGBM [7] with 511 leaves, learning rate 0.04 and 1500 trees.
2) $m_1 = 0.65\,m_0 + 0.35\,\overline{\text{CatBoost}}$ [8]: two seeds, each the mean of a depth-8 and a depth-10 model, on the 94 FD columns.
3) $m_2 = 0.65\,m_1 + 0.35\,\text{XGBoost}$ [9] (hist, depth 9, 1500 rounds).
4) **Blackout override.** About 1.9 % of the target rows lie in $(T, T{+}90\,\text{min}]$ of an official Task 2 window, where the whole mainline is dark. These rows get $0.25\,m_2 + 0.75\,\text{BSPEC}$. The specialist BSPEC is trained on simulated blackouts placed exactly as the official selector places them (onset queue first at $T{+}30$, ongoing run-starts). It has 30 timing columns and ramp flows over the gap, which are released at the normal rate inside mainline blackouts (TRACKER DIAG_RAMP).
5) **Step M.** On ordinary rows, $q \leftarrow \max(0.65q + 0.35\,q_{\text{LRTC}}, 20)$. Here $q_{\text{LRTC}}$ is an EM low-rank tensor completion of the month's own released flow cube (days × 288 slots × stations, each station normalised by its mean, 60 iterations, mean of the truncated-SVD projections of the 3 unfoldings, ranks $(24, 40, 10)$), in the spirit of HaLRTC [4].
6) **Step B.** On blackout rows, $v \leftarrow \text{clip}(0.85v + 0.15v_{\text{LRTC}}, 3, 1.05v_f)$ and $q \leftarrow \max(0.85q + 0.15q_{\text{LRTC}}, 20)$, with speed ranks $(8, 20, 20)$ and flow ranks $(8, 40, 10)$.
7) Steps R and T of Sec. V.

TABLE II
PRE-REGISTERED TASK 1 GATE ROUNDS (`ANALYSIS/T1LAB/T1DIV_LIB.PY`); $\Delta J$ AFTER RECON + TV, 8 PANEL-MONTHS.

| Rd | Candidate | $\Delta J$ ($10^{-4}$) | +/8 | Verdict |
|---|---|---|---|---|
| 1 | CatBoost FD member @0.35 | 7.1 | 8 | PASS |
| 1 | day-grid U-Net @0.15 | 0.6 | 6 | FAIL |
| 2 | XGBoost FD member @0.35 | 5.6 | 8 | PASS |
| 2 | CatBoost on old FTA columns | 0.8 | 6 | FAIL |
| 3 | XGB seed 2, ExtraTrees, ridge, log-target | $\leq$0.2 | $\leq$4 | FAIL |
| 3 | re-weight XGB to 0.6 | 1.7 | 7 | FAIL |
| 4 | flow-only LRTC member @0.35 | 5.4 | 8 | PASS |
| 4 | density interp. post-proc. (27 cfg) | $-4.1$ | 0 | FAIL |
| 4 | neighbour smoothing + FD cap | 0.6 | 6 | FAIL |
| 5 | LRTC rank mean / NNLS stack | 1.8 / 2.1 | 8 | FAIL |
| 5 | LRTC on blackout rows (blkVQ@0.15) | 1.85 | 7 | FAIL |
| 5* | ranks (24,40,10) + blkVQ@0.15 | 3.7 | 8 | post-hoc |
| 7 | kNN-day analog member | 0.5 | 6 | FAIL |
| 7 | 3-rank LRTC ensemble | 0.4 | 6 | FAIL |

The CatBoost, XGBoost and LRTC members were each added after passing the pre-registered gate: $\Delta J = +0.00071$, $+0.00056$ and $+0.00054$ (all 8/8, Table II). The blackout specialist reduced blackout-cell speed RMSE by 6 % relative to the previous ramp member ($\Delta J = +0.00044$, 8/8). The ranks of step M and the step B blend come from a *post-hoc* lead of round 5, $\Delta J = +0.00037$ [+0.00030, +0.00045], 8/8. We shipped it as K9 with an expected LB gain of about $+0.0002$ (observed $+0.00022$).

## V. TASK 3: PHYSICS THROUGH TASK 1

Task 3 cannot be submitted separately, and its LWR term is computed against organizer-held boundary fluxes. A local sensitivity study on D12_I405_S (60 days) showed how sensitive the proxy is to noise. Temporal interpolation scores 0.52, while truth plus only 1 km/h of speed noise or 1 % of flow noise falls to about 0.89. $S_{\text{LWR}}$ is therefore set mainly by the per-cell noise of the reconstructed density, not by the flux balance itself. The station-to-station conservation estimate is 10–28 times the LWR signal (TRACKER EXP_T1_LWRPROJ). We therefore improve the reconstructed density directly:

**Step R: FD-gated reconciliation.** At target rows with $v < 0.6\,v_f$ the per-lane density $k = q/(\text{lanes} \cdot v)$ is moved towards the congested FD branch implied by the predicted speed, $k_{\text{FD}}(v) = w\,k_j/(v + w)$ with $w = \text{cap}/(k_j - k_c)$. The FD parameters are averaged per link exactly as the organizer scorer does. With $r = \text{clip}(k_{\text{FD}}/k, e^{-0.5}, e^{0.5})$, we set $v' = \text{clip}(v\,r^{-0.75}, 3, 1.05v_f)$ and $q' = \max(q\,r^{0.25}, 20)$, so 75 % of the correction goes to speed. Links with implausible FD parameters ($w \notin [8, 40]$ km/h or $k_j$/lane $\notin [60, 200]$; 10 links) are excluded. A panel would also be excluded if the released congested cells disagreed with its FD (median $> 2\,\%$ or IQR outside $\pm 5\,\%$); no panel was. The step changes 4 % of state rows and gave LB $+0.00167$, the largest Task 3 gain.

**Step T: 1-D total variation.** Inside every maximal temporal run of target cells per station, the per-lane density is smoothed by a TV-L2 denoiser (300 Chambolle dual iterations [5]), anchored to the released cells just before and after the run.

The strength depends on the run's mean density (free cells $\tau = 0.0075$, gated cells 0.02). In gated cells the density change goes 75 % to speed, and in free cells all of it goes to flow. LB $+0.00026$. A scan of 294 recon/TV variants found nothing better than $+0.0002$ in proxy $J$, so both steps were frozen.

The Task 1 members also help Task 3. The flow LRTC member gains mostly through $\widetilde{S}_{\text{LWR}}$ ($+0.0043$ against $+0.00034$ in $S_{\text{state}}$; `TRACKER` EXP_T1_DIV4), because averaging with an estimator of a different kind removes part of the per-cell noise.

## VI. Task 2: queue forecasting

### A. Cell classifier and decoder

Each window is turned into one row per (link, horizon step). Features are computed only from the 60-minute history, the released row at $T$, static link attributes and priors: speed ratios and their trends, queue run lengths, neighbour and distance-to-queue features, released ramp flows at $T{-}3..T{-}1$ (ongoing only), occupancy, and train-split location/time priors $P(\text{queued} \mid \text{link}, \text{slot} \pm 2, \text{weekday class})$. A lab perturbation test checked the timing: changing any data after $T$, before $T{-}60$, or ramp flows at $\geq T$ leaves every feature unchanged (`analysis/t2lab/README.md`). Probabilities are decoded per window by an *expected-IoU* rule. Cells are sorted by $p$, and the top-$m$ are chosen to maximise $\sum_{i \leq m} p_i / (m + \sum_{i>m} p_i)$. This decoder beat fixed thresholds, morphology and per-step decoding (`TRACKER` EXP_T2_D).

### B. Onset

Post 742350 (organizer) and our own counts on the training windows (39/40 official, 96.9 % of pseudo-windows) show that every onset queue first appears at $T{+}30$. So onset is a one-step problem: *which links* queue at $T{+}30$. A CatBoost classifier (depth 8, 600 iterations, seed 0) on the step-6 rows replaced the LightGBM blend (LB $+0.0006$). For March rows ("sp4") the three prior columns are blended per cell with a *same-scenario* prior estimated from released March days strictly before the window's day, $a = n/(n+4)$ with $n$ visible days (LB $+0.00104$, the only onset change besides CatBoost that transferred).

### C. Ongoing: selection-faithful training

Training on the replica's original "current-style" pseudo-windows (Sec. III) mostly produced stable queues that the official selector never picks, because its persistence cap rejects them. Training instead on selection-faithful windows (all faithful candidates on a 15-minute grid with weight 0.3, plus faithful run-starts with weight 1.0; "F03") improved official-like OOF IoU from 0.8387 to 0.8751/0.8806 (4/4 folds) and pseudo-March by $+0.021$ (6/8 panels). On the LB it gave $+0.0047$, the largest gain after the first model. The March ongoing model ("fs4occ") is a bag of 4 LightGBM models (511 leaves, 1800 trees): two prior-free and two "identity" column sets with 2 seeds each. They use *all finite* released values as inputs, including ineligible-but-valued cells (about 7 %; LB $+0.0009$), plus 13 occupancy features ($+0.0001$).

### D. April rows and the two final picks

Train-split evidence and March evidence disagree on one question: does April share March's scenario? The release configuration says validation and private use one scenario seed, while the data README says the two months were "generated independently". On many-day pseudo-March, models fitted on train plus released-March windows clearly beat train-only models (onset $0.707 \rightarrow 0.858$ on days $\geq 15$). On held-out *training* days with March used as the wrong scenario, the same change costs $-0.13$ to $-0.14$ IoU (`TRACKER` EXP_T2_OSP_DOWNSIDE). The bet is two-sided, so we submit one file for each hypothesis. Both use only data stamped before April:

- **Pick A (tfb-k9).** April onset is a bag of 3 CatBoost models trained on train plus 215 released-March onset windows (March weights 10/20/40) with the scenario prior. April ongoing is "pvogf": a bag of prior-free, identity and full models trained on F03 rows plus released-March faithful windows ($\times 5$), with 103 additional history features. It passed its pre-registered March gate (many-day pseudo-March $+0.0219$ [+0.0117, +0.0330], 8/8). The shipped variant has a growth veto that never fired on April, so its rows equal pvogf.
- **Pick B (tfb-k10).** The same file, except for 200 April queue rows. April onset uses the train-prior CatBoost, and April ongoing uses "veto10": the train-only 103-feature bag, unless it adds more than 10 cells outside the queue at $T$, in which case fs4occ is used.

Both picks share every state, March queue and path-flow row, and both score 0.88281 on the public LB.

## VII. Task 4: ODME

For each panel and split we solve $\min_{f \geq 0} \|Af - c\|^2 + \lambda \|f - b\|^2$, $\lambda = 0.05$, as an NNLS problem on the stacked system $[A; \sqrt{\lambda} I] f = [c; \sqrt{\lambda} b]$ (`scipy.optimize.nnls` [6]). This is the organizer's reference construction (`repo/src/task4/build_task4_odme_artifacts.py`). It uses each split's own counts and prior. A $\lambda$ sweep and IRLS variants gave no gain. Our estimate is $S_{\text{ODME}} \approx 0.994$, the same for every team using this method (estimate from LB probes; `research/SCORE_GAP_ANALYSIS.md`). A validator checks path ids, OD zones, duplicates, row order and that every flow is finite and non-negative before each final submission.

## VIII. Results

Figure 2 shows the LB history and Table III the main steps. Our first own Task 2 model (LightGBM cell classifier with expected-IoU decoding) lifted the public notebook [2] from 0.80939 to 0.84357. Two probes with one condition zeroed then measured the LB IoU directly ($\Delta S = 0.3 \cdot 0.5 \cdot \text{IoU}$): onset 0.743 and ongoing 0.747 at 0.84855. Over the whole campaign, Task 2 changes account for $+0.053$ of public LB (including $+0.034$ from the first classifier), Task 1 for $+0.016$, the Task 3 post-processing for $+0.002$ and one combined step for $+0.003$ (Fig. 3). The organizer's persistence baseline scores 0.2518 on $S_{\text{queue}}$.

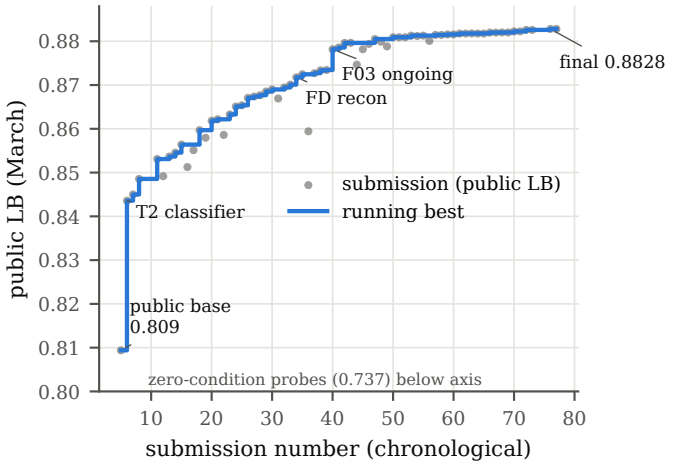



Fig. 2. Public LB of the 73 scored submissions in `all_submitted_notebooks/README.md` (from #5; the two zero-condition probes at 0.737 are off-axis) and the running best. Generated by `make_figures.py`.

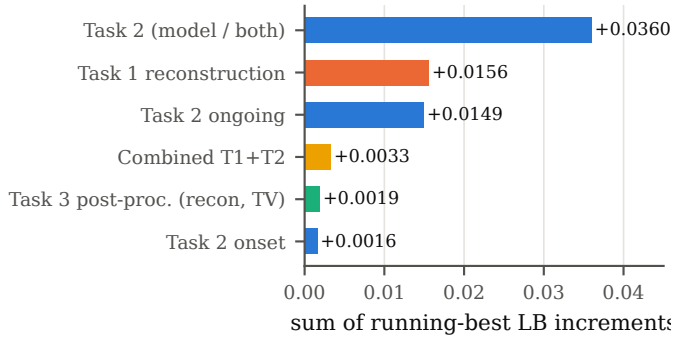



Fig. 3. Running-best public-LB increments from 0.80939 to 0.88281, summed by the task the submission changed. Increments are measured against the running best, not against a common base, so they are an attribution, not a clean ablation. Chain: `figures/lb_steps.csv`.

TABLE III
SELECTED STEPS OF THE PUBLIC-LB PROGRESSION (MARCH; EACH ROW BUILDS ON THE ROW ABOVE UNLESS NOTED IN TRACKER).

| Kernel | Task | Change | LB |
|---|---|---|---|
| exp000 | all | public notebook [2] | 0.80939 |
| probe-t2 | T2 | LightGBM cell classifier + exp.-IoU decode | 0.84357 |
| final v2 | T1 | LightGBM residual, profile-free | 0.84855 |
| swap-v5 | T2 | ongoing prior-free | 0.85309 |
| swap-v9 | T2 | weekday-class priors | 0.85639 |
| combo6 | T1 | 511 leaves, 4× rows | 0.86327 |
| combo9 | T1 | transductive rows | 0.86707 |
| c16 | T1 | bag of 8 FTA members | 0.87009 |
| c17 | T3 | FD-gated reconciliation | 0.87176 |
| c20 | T3 | 1-D TV | 0.87271 |
| e1 | T1 | + FD-feature bag | 0.87334 |
| e3 | T2 | selection-faithful ongoing (F03) | 0.87817 |
| e5 | T2 | onset same-scenario prior (sp4) | 0.87961 |
| g3 | T2 | all-finite inputs + occupancy | 0.88092 |
| h1 | T1 | blackout ramp member | 0.88129 |
| i4 | T1 | blackout specialist | 0.88176 |
| k1 | T1 | CatBoost member | 0.88198 |
| k5 | T1 | XGBoost member | 0.88225 |
| k7 | T1 | flow LRTC member | 0.88259 |
| **k9 / k10** | T1 | LRTC ranks + blackout LRTC | **0.88281** |

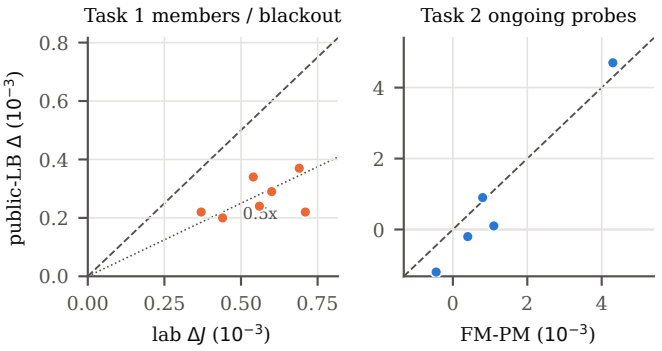



Fig. 4. Offline estimate versus realised public-LB delta. Left: Task 1 lab $\Delta J$ of the members and blackout models that were shipped (dashed: $y = x$; dotted: $y = 0.5x$). Right: Task 2 ongoing probes, FM-PM on the validation footprint. FM-PM was calibrated on these probes. Data with sources: `figdata/transfer.csv`.

**Offline-to-LB transfer.** Figure 4 (left) shows that Task 1 lab gains transferred at about 0.3–0.6×. That is expected, because the lab stand-in averages 2–4 members while production averages 17, so one extra member is diluted. The transfer was steady enough to predict the order of magnitude of each step. For example, K9 was forecast at about +0.0002 and observed at +0.00022. For Task 2 ongoing changes, FM-PM tracked the LB; plain OOF did not. For onset, no offline measure predicted the LB (Sec. IX).

**Estimated component levels.** At 0.86217 the components were estimated as $S_{\text{ODME}} \approx 0.994$, $S_{\text{FD}} \approx 0.99$, $S_{\text{state}} \approx 0.94$, $S_{\text{LWR}} \approx 0.51$ and $S_{\text{queue}} \approx 0.78$ (*estimates*, solved from LB probes; `research/SCORE_GAP_ANALYSIS.md`). Since then, +0.0095 of LB has come from Tasks 1 and 3 and +0.0111 from Task 2 (`figures/lb_steps.csv`). **[TODO: optional: final private-LB score and rank of the selected pick, once published]**

## IX. WHAT DID NOT WORK

**Onset tuning.** Table IV lists six onset changes that passed CV or pseudo-March and then lost on the LB, some heavily. An onset window has a median of about 3 queued cells, so one flipped cell moves its IoU by 0.3–0.6. Forty windows cannot separate a real gain from noise, and pseudo-March labels (observed cells only, masked histories) differ from the official noise-free labels. After these results we stopped probing onset on the LB. The pre-registered April onset extensions (location columns, March re-weighting; TRACKER EXP_T2_PV2) also failed their gates.

**Task 2 ongoing.** Several ideas failed. 103 extra history features ("ogfeat") looked good offline (official-like +0.020) but lost −0.0012 on March, because two growth calls specific to the scenario were wrong. Calibration (Platt, isotonic, power, per-step size factors) did not help, since the decoder's size had no bias. A kinematic shock-wave gate removed correct

TABLE IV
ONSET CHANGES THAT LOOKED GOOD OFFLINE BUT LOST ON THE PUBLIC LB (TRACKER).

| Change | Offline evidence | LB Δ |
|---|---|---|
| 30/70 main/prior-free blend | local onset worse | −0.0039 |
| 50/50 CatBoost / LGB | pseudo-March +0.018 | −0.0021 |
| CatBoost bag + stage-2 stacker | CV +0.0064, 4/4 folds | −0.0130 |
| leave-one-day-out priors | pseudo-March +0.018..0.028 | −0.0050 |
| early-month mixture prior | forecast +0.004 | −0.0014 |
| all-finite inputs for onset | train OOF +0.004..0.006 | −0.0011 |

growth cells. Reversing link order in W/S panels gave −0.003. A regime switch for windows that start mid-run (−0.0002 LB) did not help either. Raw space-time patches, a vehicle-count / jam-length estimate, occupancy on the April model and blending in 30 % persistence (−0.0018 LB) all failed.

**Task 1 / 3.** Other approaches also failed. A Kalman/RTS smoother (−0.019 composite), interpolating per-lane flow instead of total flow, a density model $q = \hat{k}\hat{v}$, per-regime models and an L1 density model all lost. A day-grid U-Net (raw speed RMSE 2.5 against 1.7 for LightGBM), ExtraTrees, ridge and kNN-day members failed too. Density time-interpolation post-processing from public notebooks was negative in all 27 settings (our features already contain the interpolation), and so was a conservation-aware projection, whose best variant reached only +0.00006. The LWR projection failed for a structural reason: on these panels the station-to-station flux estimate is noisier than the signal.

**Task 4.** A $\lambda$ sweep and IRLS gave no gain over the organizer's NNLS.

## X. REPRODUCIBILITY

**Entry script.** `consolidation/run_all.py` rebuilds either pick from the competition release with one command. Each stage is the original Kaggle kernel, vendored with only its I/O lines changed (each marked `# [TFB]`, which a static test enforces). A template test checks that every rendered Task 1 member part equals its original source, compared as an AST. **[TODO: final command line for pick A3 = tfb-k9 and B3 = tfb-k10, e.g. `python run_all.py -data-root <kaggle_public> -work-dir ./work -pick A3 -recipe full`; confirm against the updated `consolidation/README.md` / `LINEAGE_K9.md`] [TODO: code location submitted to the organizers (GitHub URL or zip name)]**

**Compute.** All members ran on Kaggle CPU kernels (4 vCPU, about 30 GB RAM). A Task 1 LightGBM member takes 4.6–8.8 h over its 2–3 panel parts, with a peak of 28.0 GB. The CatBoost members take 15.7 h per seed over 4 parts and the XGBoost member 4.2 h (`consolidation/lineage_K9.json`). The previous pick (tfb-j4) was measured at 107–111 Kaggle CPU-hours in the full recipe. Adding the k generation gives an *estimated* 143–147 CPU-hours for tfb-k9. On one 8-core, 32 GB Linux machine the earlier full recipe was estimated at 63–68 h of wall time. **[TODO: measured end-to-end wall time and peak RAM of the consolidated A3 / B3 run] [TODO: runtime of the tfb-k9 kernel itself (estimated 16–20 min; tfb-k7 took 771 s)]**

**Environment.** Python 3.13 Kaggle image (digest in `consolidation/ENVIRONMENT.md`). The parity runs used numpy 2.1.3, pandas 2.3.3, scipy 1.16.3, LightGBM 4.6.0, CatBoost 1.2.10 and pyarrow 23.0.1. **[TODO: pin the exact package versions of the reference image (pip freeze)]**

**Determinism.** Seeds, thread counts and operation order are those of the original kernels. Each stage runs in its own process with `PYTHONHASHSEED=0` and 4 OMP/BLAS threads. CatBoost reproduced exactly across images (agreement 1.000000). Light stages (onset, Task 1 mix, recon/TV, assembly, Task 4) matched the Kaggle outputs exactly. The original LightGBM kernels did not fix LightGBM's histogram construction, so reruns of the Task 2 ongoing models differ in a few dozen of 87,000 cells. The private-score bound for this is about 0.0012, within the organizers' 1 % tolerance (`consolidation/README.md`). The consolidated stages therefore run LightGBM with `deterministic=True`, `force_row_wise=True`. **[TODO: Kaggle parity of the consolidated A3 / B3 output versus the submitted tfb-k9 / tfb-k10 files: queue-row agreement, state RMS per split, path-flow equality, and the resulting score difference]**

## XI. LIMITATIONS AND CONCLUSION

Only 40 windows per condition are scored per month, so Task 2 effects below about 0.002 cannot be seen on the LB, and our onset model is still tuned mainly offline. The April picks hedge one structural uncertainty (scenario sharing) and cannot resolve it. Most of the compute goes into a Task 1 ensemble whose last members add +0.0002 each. Our main lessons apply beyond this competition. Reproduce the evaluation's *selection* process before training on it. Physics enters this score mainly through noise in the reconstructed density, so physics-aware post-processing should target that noise. Pre-registered gates with controls and a measured offline-to-LB transfer ratio made a 73-submission campaign much less prone to fitting LB noise.


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We thank the TrafficFlowBench organizers for the benchmark and their forum answers. Our pipeline started from the public notebook of Sōsuke Aizen [2]. The 1-D TV idea follows a public notebook by Hisernberg. The month-level low-rank completion idea was suggested by a public notebook by plarmuseau, and the CatBoost member settings follow a public notebook by anubisyy. All code in our final pipeline was written or re-implemented by the team, and no external data was used.


## REFERENCES


[1] TrafficFlowBench organizers, "TrafficFlowBench: a four-task freeway traffic benchmark for the 2026 IEEE Big Data Cup," repository README and docs (`repo/`), and Kaggle competition "2026 IEEE Big Data – Traffic Flow Bench," 2026.

[2] S. Aizen, "[0.809 LB] Open-sourcing our physics-consistent 4-task pipeline," Kaggle notebook and forum post 742980, 2026.

[3] M. J. Lighthill and G. B. Whitham, "On kinematic waves II: a theory of traffic flow on long crowded roads," *Proc. R. Soc. A*, vol. 229, pp. 317–345, 1955; P. I. Richards, "Shock waves on the highway," *Oper. Res.*, vol. 4, pp. 42–51, 1956.

[4] J. Liu, P. Musialski, P. Wonka and J. Ye, "Tensor completion for estimating missing values in visual data," *IEEE TPAMI*, vol. 35, no. 1, pp. 208–220, 2013.
[5] A. Chambolle, "An algorithm for total variation minimization and applications," *J. Math. Imaging Vis.*, vol. 20, pp. 89–97, 2004.
[6] C. L. Lawson and R. J. Hanson, *Solving Least Squares Problems*. Prentice-Hall, 1974.
[7] G. Ke et al., "LightGBM: a highly efficient gradient boosting decision tree," in *NeurIPS*, 2017.
[8] L. Prokhorenkova et al., "CatBoost: unbiased boosting with categorical features," in *NeurIPS*, 2018.
[9] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *KDD*, 2016.